CHAPTER 9

DEFORMATION OF SOLIDS

Key idea: within the limit of proportionality, stress is proportional to strain.

1 STRESS & STRAIN

$$\sigma = \frac{F}{A} \qquad \varepsilon = \frac{x}{L_0}$$

Stress normal force per unit area, Pa (N m^{-2})

Strain extension ÷ original length, **no unit**

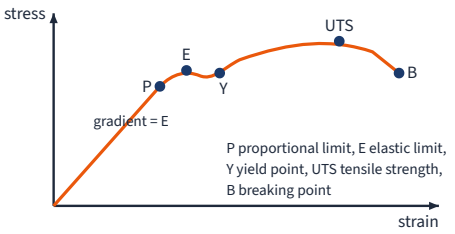
Wire of diameter d: $A = \dfrac{\pi d^2}{4}$

2 YOUNG'S MODULUS

$$E = \frac{\sigma}{\varepsilon} = \frac{FL_0}{Ax}$$

- Gradient of the linear part of the σ–ε graph
- Unit Pa; a property of the **material**, not of the sample

3 STRESS–STRAIN GRAPH



P limit of proportionality
E elastic limit
Y yield point: plastic flow
UTS maximum stress
B breaking point

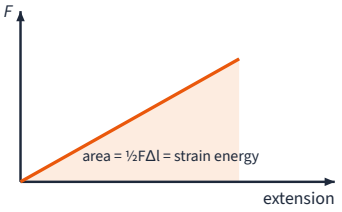
4 HOOKE'S LAW

$$F = kx \qquad k = \frac{EA}{L_0}$$

Extension ∝ force, provided the limit of proportionality is not exceeded.

- Elastic: returns to original length
- Plastic: permanent extension remains

5 ELASTIC POTENTIAL ENERGY



$$U = \tfrac{1}{2}Fx = \tfrac{1}{2}kx^2$$

$$\frac{U}{V} = \tfrac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2E}$$

Energy = **area** under the F–x graph.

6 TYPES OF MATERIAL

Ductile large plastic region before breaking (copper, mild steel)

Brittle breaks just after the elastic region (glass, cast iron)

Rubber large strain; unloading curve below loading: area between = heat lost (hysteresis)

7 UNIT CHECK

Quantity	Formula	Unit
Stress	F/A	Pa
Strain	x/L_0	–
Young modulus	σ/ε	Pa
Force constant	EA/L_0	N m^{-1}
Energy density	$\frac{1}{2}\sigma\varepsilon$	J m^{-3}

8 COMMON SECTION A TRAPS

- Using the diameter as the radius in $A = \pi r^2$.
- Forgetting mm → m for extension and diameter.
- Using $U = Fx$ instead of $\frac{1}{2}Fx$.
- Thinking a thicker wire of the same metal has a larger E (it has a larger k).
- Applying $\frac{1}{2}Fx$ beyond the limit of proportionality.

EXAM TIPS

- ✓Find E from the gradient of the straight part of the graph.
- ✓Give strain as a pure number, e.g. 2.0×10^{-3}.
- ✓Use ratios when two wires of the same material are compared.
- ✓Area under a curve: count squares or use trapezia.